

Youssef Elmougy

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Portland, OR 97006, United States

EDUCATION

- **Ph.D. in Computer Science** Aug 2022 - May 2025
Georgia Institute of Technology Atlanta, GA
 - Research concentrated in HPC, Systems, and AI/DL. Worked in the Habanero Extreme Scale Software Research Lab. Advisor: Vivek Sarkar.
 - Thesis: Scalable Asynchronous Actor-based Approaches for Distributed-Memory Parallel Applications.
- **M.S. in Computer Science** Jan 2022 - Dec 2022
Georgia Institute of Technology Atlanta, GA
 - Specialization in High Performance Computing. GPA: 3.6/4.0. IEEE-HKN Student Member.
- **B.S. in Computer Science** Aug 2017 - Dec 2021
Georgia Institute of Technology Atlanta, GA
 - Specialization in Artificial Intelligence and Computer Modelling. GPA: 3.8/4.0. IEEE-HKN Student Member. Graduated with Highest Honors.
 - Transferred from Hofstra University (attended Aug 2017 - May 2020) with 4.0/4.0 GPA as a Presidential Scholarship Recipient.

EXPERIENCE

- **Senior Software Engineer** May 2025 - PRESENT
NVIDIA Portland, OR
 - Working within the Advanced Technology Group (ATG) on the cuLitho project.
 - Design and implement high-performance algorithms to GPU-accelerate computational lithography and the manufacturing process of semiconductors by orders of magnitude over current CPU-based methods.
 - Collaborate with external partners to deploy cuLitho into their semiconductor production environment, delivering 40x performance speedup, 3x-5x productivity improvement, and cost savings (500 NVIDIA Hopper GPU systems vs 40K CPU systems).
- **Graduate Research Assistant** May 2022 - May 2025
Habanero Extreme Scale Software Research Lab, Georgia Institute of Technology Atlanta, GA
 - Increased resiliency and performance of the HCLib Actor-based runtime system by extending automatic communication termination protocols, distributed graph generation, and multithread execution.
 - Built large-scale distributed graph algorithms, including triangle centrality, jaccard index, page rank, pattern matching, genome comparisons, internet network topology analysis, deep learning, and GNN.
 - Implemented a distributed and shared-memory parallel Actor-based runtime system for cloud computing, allowing for HPC on the Cloud with low latency.
 - Optimized and fine tuned the runtime system using an architecture-aware approach, including evaluating intra-node core, socket, NUMA, and software-level buffer bindings.
- **Graduate Research Intern** May 2024 - Aug 2024
NVIDIA - NV Research Santa Clara, CA
 - Worked within the Programming Systems and Applications Research Group to analyze the performance limitations of NVIDIA's state-of-the-art libraries for graph analytics. Mentor: Michael Garland.
 - Designed and implemented new scalable algorithmic approaches to improve performance and reduce code complexity of these graph libraries using the CUDASTF task-based programming system.
 - Collaborated with NVIDIA product teams to guide research & development of algorithms and software.
- **HPC Teaching Assistant** Jan 2024 - May 2024
Georgia Institute of Technology Atlanta, GA
 - TA for CSE 6220 - Introduction to High Performance Computing.

- Engaged with students on topics of theoretical complexities, parallel and distributed computing, and communication efficient algorithms through weekly office hours. Prepared MPI-based HPC projects.
- **Graduate Research Intern** May 2023 - Aug 2023
Lawrence Berkeley National Lab San Francisco, CA
 - Worked within the Performance and Algorithms Research Lab on hybrid communication techniques and increasing fault tolerance of distributed learning for deep learning workflows. Mentor: Khaled Ibrahim.
 - Built a hybrid AllReduce and Parameter Server approach to parameter distribution/update and collective communication for distributed training using PyTorch DDP and RPC.
- **Robotics Teaching Assistant** Aug 2021 - May 2022
Georgia Institute of Technology Atlanta, GA
 - TA for CS 3630 - Introduction to Perception and Robotics.
 - Engaged with students on topics of robotics planning, control and localization through weekly office hours. Prepared Cozmo and Vector robots for Labs.
- **Research Assistant** Aug 2020 - Dec 2021
Automated Algorithm Design, Georgia Institute of Technology Atlanta, GA
 - Worked within Stocks subteam of AAD to alter the use of machine learning techniques in developing hybrid algorithms for stock price prediction. Mentor: Jason Zutty.
 - Programmed stock trading related primitives, objective functions, and genetic programming frameworks built on top of EMAD.
- **Research Assistant** May 2019 - May 2020
Hofstra University Hempstead, NY
 - Worked on systems and cloud infrastructure research. Mentor: Jianchen Shan.
 - Research on diagnosing and optimizing the performance interference caused by CPU sharing in multi-tenant GPU clouds. Presented at ASPiRe Symposium '19, published paper in IPCCC '21.
- **SEAS IT Technician** May 2019 - May 2020
EdTech, Hofstra University Hempstead, NY
 - Provide technical support to faculty members in the DeMatteis School of Engineering and Applied Science.
 - Primary support includes specialized software installation and configuration, hardware setup, and classroom technology support.
- **Data Analytics and Web Developer Intern** Jan 2019 - May 2019
FORKaia Irvine, CA
 - Gathered specifications based on technical needs. Defined a data analysis process, and identified patterns and trends in datasets. Worked on the apps: Namebeat, Heirgraphics, Aura App.

PUBLICATIONS [🎓]

C=CONFERENCE, J=JOURNAL, P=POSTER, S=UNDER SUBMISSION, G=GITHUB, A=AWARD

- [S.1] Youssef Elmougy, Akihiro Hayashi, Vivek Sarkar. (2025). **HybridFusion: A Parameter Update Framework for Scalable and Efficient Distributed Deep Learning**. Under submission.
- [C.1] Youssef Elmougy, Akihiro Hayashi, Vivek Sarkar. (2025). **Divide, Conquer, and Match: A Distributed and Asynchronous Approach for Subgraph Isomorphism**. In *GrAPL at IEEE IPDPS*.
- [C.2] Youssef Elmougy, Nirjhar Deb, Akihiro Hayashi, Vivek Sarkar. (2025). **Enhancing Productivity and Performance of HCLib-Actor with Efficient Task Termination**. In *APDCM at IEEE IPDPS*.
- [C.3] Youssef Elmougy, Shubhendra Singhal, Akihiro Hayashi, Vivek Sarkar. (2025). **ActorISx: Exploiting Asynchrony for Scalable High-Performance Integer Sort**. In *IEEE/ACM CCGRID*.
- [C.4] Aniruddha Mysore, Youssef Elmougy, Akihiro Hayashi. (2024). **On the Cloud We Can't Wait: Asynchronous Actors Perform Even Better on the Cloud**. In *VIVEKFEST Symposium at SPLASH*.
- [C.5] Akihiro Hayashi, Shubhendra Singhal, Youssef Elmougy, Jiawei Yang. (2024). **Enabling User-level Asynchronous Tasking in the FA-BSP Model - Case Study: Distributed Triangle Counting**. In *VIVEKFEST Symposium at SPLASH*.
- [C.6] [G] Youssef Elmougy, Akihiro Hayashi, Vivek Sarkar. (2024). **Asynchronous Distributed Actor-based Approach to Jaccard Similarity for Genome Comparisons**. In *IEEE ISC HPC*.

- [C.7]🏆 Youssef Elmougy, Akihiro Hayashi, Vivek Sarkar. (2024). **A Distributed, Asynchronous Algorithm for Large-Scale Internet Network Topology Analysis**. In *IEEE/ACM CCGRID*. Winner of TCSC SCALE Award.
- [C.8]🏆 Youssef Elmougy, Ling Liu. (2023). **Demystifying Fraudulent Transactions and Illicit Nodes in the Bitcoin Network for Financial Forensics**. In *ACM SIGKDD*.
- [C.9]🏆 Youssef Elmougy, Akihiro Hayashi, Vivek Sarkar. (2023). **Highly Scalable Large-Scale Asynchronous Graph Processing using Actors**. In *IEEE/ACM CCGRID*. Winner of TCSC SCALE Award.
- [C.10] Youssef Elmougy, Weiwei Jia, Xiaoning Ding, Jianchen Shan. (2021). **Diagnosing the Interference on CPU-GPU Synchronization Caused by CPU Sharing in Multi-Tenant GPU Clouds**. In *IEEE IPCCC*.
- [C.11] Youssef Elmougy, Oliver Manzi. (2021). **Anomaly Detection on Bitcoin, Ethereum Networks Using GPU-accelerated Machine Learning Methods**. In *IEEE ICCTA*.
- [J.1]🏆 Sri Raj Paul, Akihiro Hayashi, Kun Chen, Youssef Elmougy, Vivek Sarkar. (2023). **A Fine-grained Asynchronous Bulk Synchronous Parallelism Model for PGAS Applications**. In *Journal of Computational Science*.
- [P.1] Aniruddha Mysore, Kaushik Ravichandran, Youssef Elmougy, Akihiro Hayashi, Vivek Sarkar. (2023). **Accelerating Actor-based Distributed Triangle Counting**. In *IEEE/ACM SC*.

REVIEWER AND CONTRIBUTOR

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|--|-------------|
| • ACM Transactions on Knowledge Discovery from Data (reviewer) | 2025 |
| • IEEE Transactions on Big Data (reviewer) | 2025 |
| • OpenSHMEM Application Programming Interface, Version 1.6 (contributor) | 2024 |
| • ACM Transactions on Internet Technology (reviewer) | 2022 - 2023 |
| • IEEE Cloud Summit 2021 (reviewer) | 2021 |

HONORS AND AWARDS

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| • International Scalable Computing Challenge Award (SCALE 2024) 🏆
<i>IEEE TCSC (Technical Committee on Scalable Computing) at the IEEE/ACM CCGrid Conference</i> | 2024 |
| • Innovative Use of High Performance Computing Award 🏆
<i>The National Energy Research Scientific Computing Center (NERSC) and the U.S. Department of Energy (DOE) Office of Science</i> | 2023 |
| • Inspiration Award
<i>At the 2023 Monte Jade Innovation Competition for the "Streaming Digital Innovation into Services with Blockchain" project</i> | 2023 |
| • International Scalable Computing Challenge Award (SCALE 2023) 🏆
<i>IEEE TCSC (Technical Committee on Scalable Computing) at the IEEE/ACM CCGrid Conference</i> | 2023 |
| • Microsoft Azure Grant for \$10,500
<i>IDEaS Cloud Hub at Georgia Tech</i> | 2023 |
| • Phi Beta Kappa Book Award
<i>Phi Beta Kappa Association of New York</i> | 2019 |
| • Presidential Scholarship recipient
<i>Hofstra University</i> | 2017 - 2020 |

SKILLS

Programming: C++/C/C#, Python, CUDA, Java, FLEXSIM, MATLAB, HTML/CSS, ROS, Coq, GIT
Libraries: MPI, OpenSHMEM, UPC, Conveyors, Slurm, cuGraph, CUB, CUDASTF, cuLitho, Thrust, NCCL, CCCL
ML Frameworks: PyTorch, TensorFlow, HuggingFace, Scikit Learn
Virtualization: Docker, Singularity, KVM, Linux
Cloud: AWS, GCP, Azure
Languages: English, Arabic, French